AI-Based Resume Screening System

In [1]:
```
print("Hello Omkar")
```

```
Hello Omkar
```

In [2]:
```
!pip install pandas scikit-learn
```

```
Requirement already satisfied: pandas in .\appdata\local\programs\python\python31
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  Downloading scikit_learn-1.7.2-cp310-cp310-win_amd64.whl.metadata (11 kB)
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ython310\lib\site-packages (from pandas) (2.2.6)
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thon310\lib\site-packages (from scikit-learn) (1.15.3)
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Collecting threadpoolctl>=3.1.0 (from scikit-learn)
  Downloading threadpoolctl-3.6.0-py3-none-any.whl.metadata (13 kB)
Requirement already satisfied: six>=1.5 in .\appdata\local\programs\python\python
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Downloading joblib-1.5.3-py3-none-any.whl (309 kB)
Downloading threadpoolctl-3.6.0-py3-none-any.whl (18 kB)
Installing collected packages: threadpoolctl, joblib, scikit-learn

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Successfully installed joblib-1.5.3 scikit-learn-1.7.2 threadpoolctl-3.6.0
```

```
[notice] A new release of pip is available: 26.1.2 -> 26.2.1
[notice] To update, run: python.exe -m pip install --upgrade pip
```

1. Creating Job Description and Sample Resumes

In [3]:
```
job_description = """
Data Analyst with skills in Python, SQL, Excel, Tableau, and Data Visualization.
"""

resume_1 = """
I know Python, SQL, Excel and Data Visualization.
"""

resume_2 = """
I have experience in Java, HTML, CSS and Web Development.
"""

print("Job Description:")
print(job_description)

print("\nResume 1:")
print(resume_1)

print("\nResume 2:")
print(resume_2)
```

```
Job Description:

Data Analyst with skills in Python, SQL, Excel, Tableau, and Data Visualization.


Resume 1:

I know Python, SQL, Excel and Data Visualization.


Resume 2:

I have experience in Java, HTML, CSS and Web Development.
```

2. Resume Matching Using TF-IDF and Cosine Similarity

In [5]:
```
from sklearn.feature_extraction.text import TfidfVectorizer
from sklearn.metrics.pairwise import cosine_similarity

# Put all text together
documents = [job_description, resume_1, resume_2]

# Convert text into numbers
vectorizer = TfidfVectorizer()
tfidf_matrix = vectorizer.fit_transform(documents)

# Compare resumes with job description
similarity_scores = cosine_similarity(
    tfidf_matrix[0:1],
```

```
    tfidf_matrix[1:]
)

# Print results
print("Resume 1 Match Score:", round(similarity_scores[0][0] * 100, 2), "%")
print("Resume 2 Match Score:", round(similarity_scores[0][1] * 100, 2), "%")
```

```
Resume 1 Match Score: 60.0 %
Resume 2 Match Score: 10.65 %
```

3. Candidate Ranking

In [6]:
```
import pandas as pd

results = pd.DataFrame({
    "Candidate": ["Resume 1", "Resume 2"],
    "Match Score (%)": [
        round(similarity_scores[0][0] * 100, 2),
        round(similarity_scores[0][1] * 100, 2)
    ]
})

# Sort from highest score to lowest
results = results.sort_values(
    by="Match Score (%)",
    ascending=False
)

results
```

Out[6]:

	Candidate	Match Score (%)
0	Resume 1	60.00
1	Resume 2	10.65

In [8]:
```
results = pd.DataFrame({
    "Candidate": ["Omkar Gadhe", "Rahul kale"],
    "Match Score (%)": [
        round(similarity_scores[0][0] * 100, 2),
        round(similarity_scores[0][1] * 100, 2)
    ]
})

results = results.sort_values(
    by="Match Score (%)",
    ascending=False
).reset_index(drop=True)

# Add ranking
results.index = results.index + 1
results.index.name = "Rank"

results
```

Out[8]:

Rank	Candidate	Match Score (%)
1	Omkar Gadhe	60.00
2	Rahul kale	10.65

In [9]:
```
resume_3 = """
Data Analyst with experience in Python, SQL, Tableau, Excel,
Machine Learning and Data Visualization.
"""

print(resume_3)
```

```
Data Analyst with experience in Python, SQL, Tableau, Excel,
Machine Learning and Data Visualization.
```

In [10]:
```
from sklearn.feature_extraction.text import TfidfVectorizer
from sklearn.metrics.pairwise import cosine_similarity

# Add job description and all 3 resumes
documents = [
    job_description,
    resume_1,
    resume_2,
    resume_3
]

# Convert text into numbers
vectorizer = TfidfVectorizer()
tfidf_matrix = vectorizer.fit_transform(documents)

# Compare all resumes with the job description
similarity_scores = cosine_similarity(
    tfidf_matrix[0:1],
    tfidf_matrix[1:]
)

# Show scores
print("Resume 1 Match:", round(similarity_scores[0][0] * 100, 2), "%")
print("Resume 2 Match:", round(similarity_scores[0][1] * 100, 2), "%")
print("Resume 3 Match:", round(similarity_scores[0][2] * 100, 2), "%")
```

```
Resume 1 Match: 57.25 %
Resume 2 Match: 9.64 %
Resume 3 Match: 76.66 %
```

In [11]:
```
Resume 1 Match: ...
Resume 2 Match: ...
Resume 3 Match: ...
```

```
  Cell In[11], line 1
    Resume 1 Match: ...
           ^
SyntaxError: invalid syntax
```

In [12]:
```
Resume 1 Match: ...
Resume 2 Match: ...
```

```
Resume 3 Match: ...
```

```
  Cell In[12], line 1
    Resume 1 Match: ...
           ^
SyntaxError: invalid syntax
```

In [14]:
```
results = pd.DataFrame({
    "Candidate": ["Om jatti", "Priya Patil", "Karan shelke"],
    "Match Score (%)": [
        round(similarity_scores[0][0] * 100, 2),
        round(similarity_scores[0][1] * 100, 2),
        round(similarity_scores[0][2] * 100, 2)
    ]
})

# Sort candidates from highest match to lowest
results = results.sort_values(
    by="Match Score (%)",
    ascending=False
).reset_index(drop=True)

# Add rank numbers
results.index = results.index + 1
results.index.name = "Rank"

results
```

Out[14]:

Rank	Candidate	Match Score (%)
1	Karan shelke	76.66
2	Om jatti	57.25
3	Priya Patil	9.64

In [16]:
```
"Om jatti"
"Priya Patil"
"Karan shelke"
```

Out[16]:
```
'Karan shelke'
```

In [17]:
```
#Job Description Skills:
#Python, SQL, Excel, Tableau, Data Visualization
```

```
  Cell In[17], line 1
    Python, SQL, Excel, Tableau, Data Visualization
                                   ^
SyntaxError: invalid syntax
```

4. Skill Extraction

In [18]:
```
# List of skills we want to check
skills = [
    "python",
    "sql",
```

```
    "excel",
    "tableau",
    "machine learning",
    "data visualization"
]

def extract_skills(text):
    text = text.lower()

    found_skills = []

    for skill in skills:
        if skill in text:
            found_skills.append(skill)

    return found_skills


# Check skills for each resume
print("Resume 1 Skills:", extract_skills(resume_1))
print("Resume 2 Skills:", extract_skills(resume_2))
print("Resume 3 Skills:", extract_skills(resume_3))
```

```
Resume 1 Skills: ['python', 'sql', 'excel', 'data visualization']
Resume 2 Skills: []
Resume 3 Skills: ['python', 'sql', 'excel', 'tableau', 'machine learning', 'data
visualization']
```

5. Missing Skills Detection

In [19]:
```
job_skills = extract_skills(job_description)

def find_missing_skills(resume_text):
    resume_skills = extract_skills(resume_text)

    missing_skills = []

    for skill in job_skills:
        if skill not in resume_skills:
            missing_skills.append(skill)

    return missing_skills


print("Job Skills:", job_skills)

print("\nResume 1 Missing Skills:", find_missing_skills(resume_1))
print("Resume 2 Missing Skills:", find_missing_skills(resume_2))
print("Resume 3 Missing Skills:", find_missing_skills(resume_3))
```

```
Job Skills: ['python', 'sql', 'excel', 'tableau', 'data visualization']

Resume 1 Missing Skills: ['tableau']
Resume 2 Missing Skills: ['python', 'sql', 'excel', 'tableau', 'data visualizatio
n']
Resume 3 Missing Skills: []
```

In [20]:
```
job_skills = extract_skills(job_description)
```

```
def find_missing_skills(resume_text):
    resume_skills = extract_skills(resume_text)

    missing_skills = []

    for skill in job_skills:
        if skill not in resume_skills:
            missing_skills.append(skill)

    return missing_skills


print("Job Skills:", job_skills)

print("\nResume 1 Missing Skills:", find_missing_skills(resume_1))
print("Resume 2 Missing Skills:", find_missing_skills(resume_2))
print("Resume 3 Missing Skills:", find_missing_skills(resume_3))
```

```
Job Skills: ['python', 'sql', 'excel', 'tableau', 'data visualization']

Resume 1 Missing Skills: ['tableau']
Resume 2 Missing Skills: ['python', 'sql', 'excel', 'tableau', 'data visualizatio
n']
Resume 3 Missing Skills: []
```

6. Final Candidate Results

In [22]:

```
final_results = pd.DataFrame({
    "Candidate": ["Om jatti", "Priya Patil", "Karan shelke"],

    "Match Score (%)": [
        round(similarity_scores[0][0] * 100, 2),
        round(similarity_scores[0][1] * 100, 2),
        round(similarity_scores[0][2] * 100, 2)
    ],

    "Skills Found": [
        ", ".join(extract_skills(resume_1)),
        ", ".join(extract_skills(resume_2)),
        ", ".join(extract_skills(resume_3))
    ],

    "Missing Skills": [
        ", ".join(find_missing_skills(resume_1)),
        ", ".join(find_missing_skills(resume_2)),
        ", ".join(find_missing_skills(resume_3))
    ]
})

# Sort by highest match score
final_results = final_results.sort_values(
    by="Match Score (%)",
    ascending=False
).reset_index(drop=True)

# Add ranking
final_results.index = final_results.index + 1
final_results.index.name = "Rank"
```

```
final_results
```

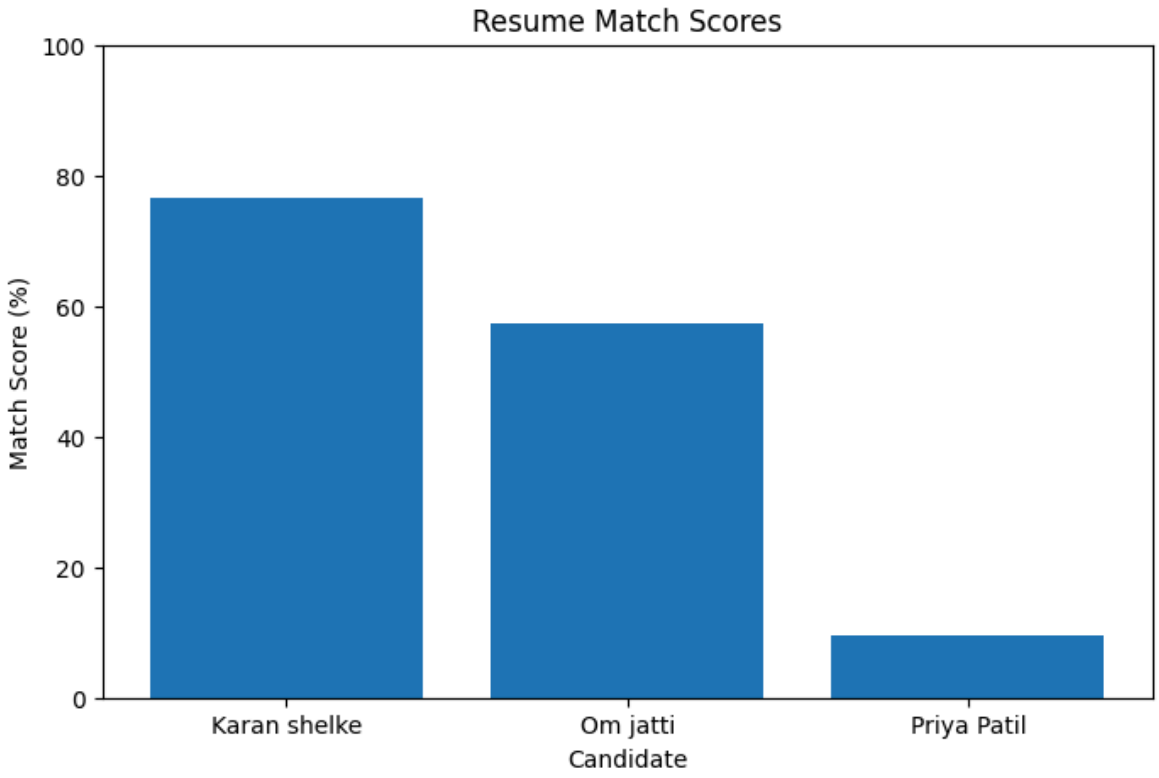
Out[22]:

Rank	Candidate	Match Score (%)	Skills Found	Missing Skills
1	Karan shelke	76.66	python, sql, excel, tableau, machine learning,...	
2	Om jatti	57.25	python, sql, excel, data visualization	tableau
3	Priya Patil	9.64		python, sql, excel, tableau, data visualization

7. Match Score Visualization

In [23]:

```
import matplotlib.pyplot as plt

plt.figure(figsize=(8, 5))

plt.bar(
    final_results["Candidate"],
    final_results["Match Score (%)"]
)

plt.title("Resume Match Scores")
plt.xlabel("Candidate")
plt.ylabel("Match Score (%)")

plt.ylim(0, 100)

plt.show()
```



8. Candidate Shortlisting

In [24]:
```
def get_status(score):
    if score >= 50:
        return "Shortlisted"
    else:
        return "Rejected"


final_results["Status"] = final_results["Match Score (%)"].apply(get_status)

final_results
```

Out[24]:

Rank	Candidate	Match Score (%)	Skills Found	Missing Skills	Status
1	Karan shelke	76.66	python, sql, excel, tableau, machine learning,...		Shortlisted
2	Om jatti	57.25	python, sql, excel, data visualization	tableau	Shortlisted
3	Priya Patil	9.64		python, sql, excel, tableau, data visualization	Rejected

Conclusion

This project successfully demonstrates an AI-Based Resume Screening System. The system compares candidate resumes with a job description using TF-IDF and Cosine Similarity.

Candidates are ranked based on their match scores. The system also identifies skills found, missing skills, and automatically classifies candidates as Shortlisted or Rejected.

In [ ]: